

Delta Quant Experiments

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Outline

- Analysis metrics on unrotated, walsh-hadamard rotated and block-diagonal rotated activations
- QuaRot and DuQuant Results using 1 anchor
- New Calibration method on DuQuant
- Effect of uniform reanchoring on results
- Effect of block reanchoring on results
- Effect of clipping on results

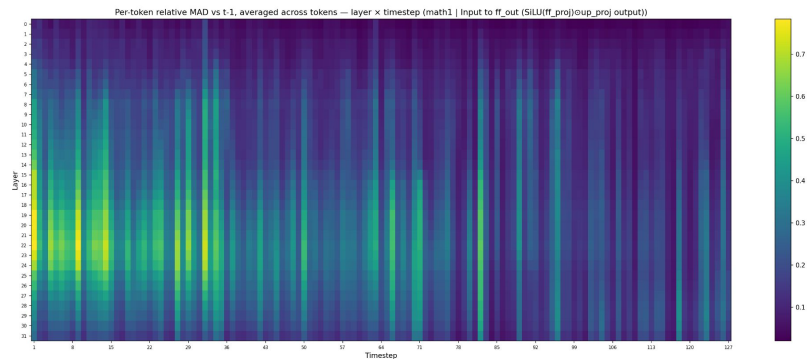
Analysis Plots

Three changes were made to the analysis plots from last time

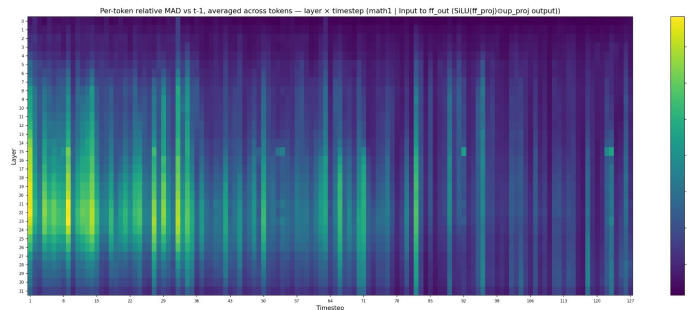
- Per-token metrics instead of per tensor
- Metrics were calculated relative to the current timestep instead of previous timestep
- Added SNR (in dB) metrics that calculate how much delta-quantizing ($x_t - x_{t-1}$) achieves when reconstructing x_t , versus directly quantizing x_t
-

Relative Mean Absolute Difference: ff_out_in activations

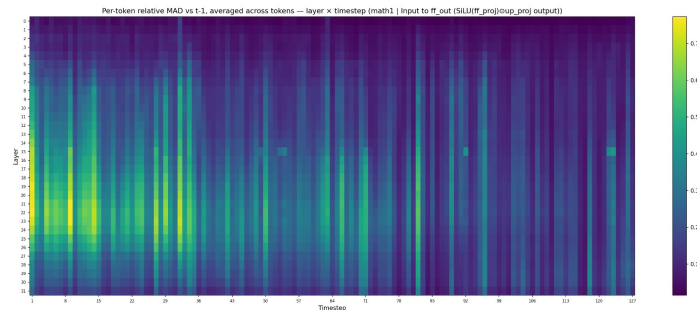
Unrotated Activations



Walsh-Hadamard Rotated (QuaRot)

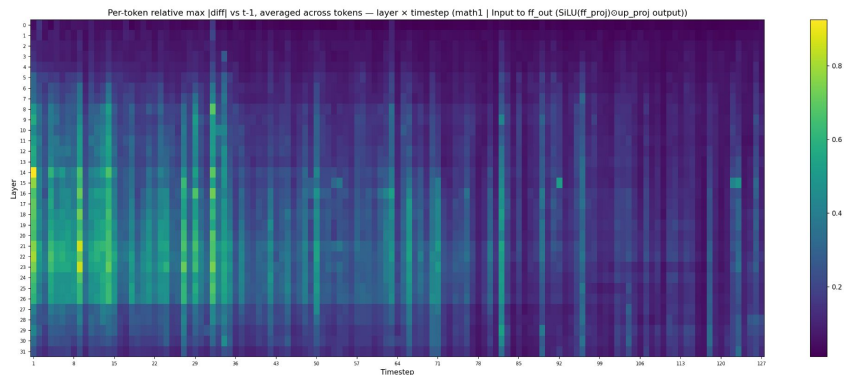


Block-Diagonal Rotated + Permuted (DuQuant)

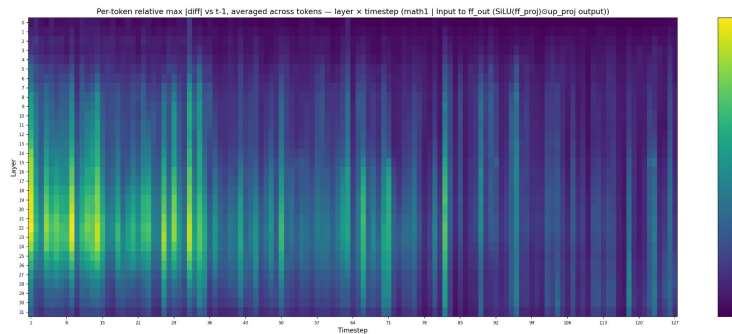


Relative Max Absolute Difference: ff_out_in activations

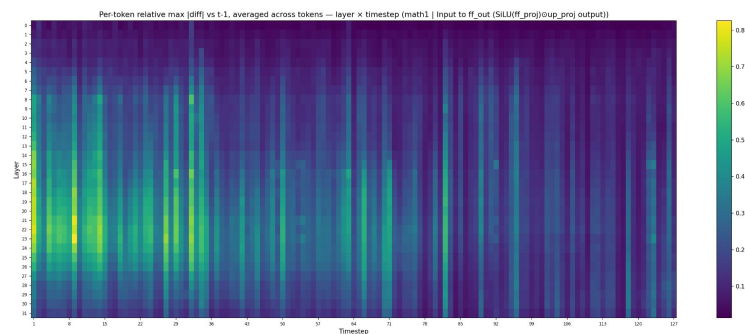
Unrotated Activations



Walsh-Hadamard Rotated (QuaRot)

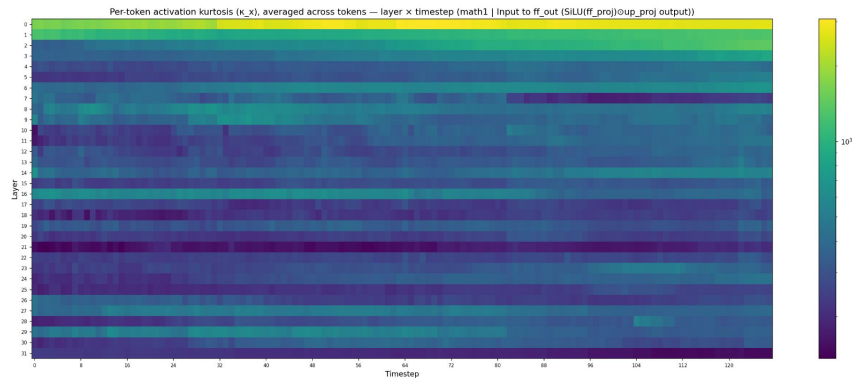


Block-Diagonal Rotated + Permuted (DuQuant)

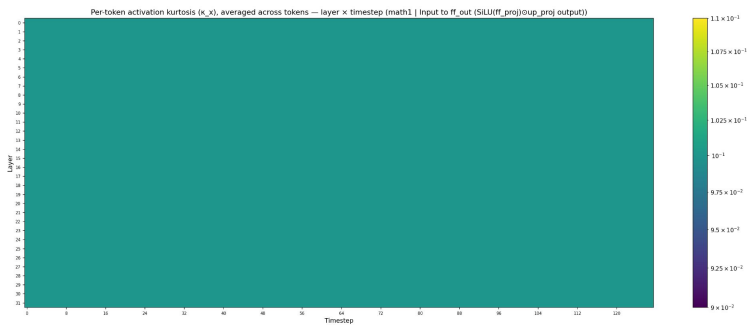


Activation Kurtosis : ff_out_in activations

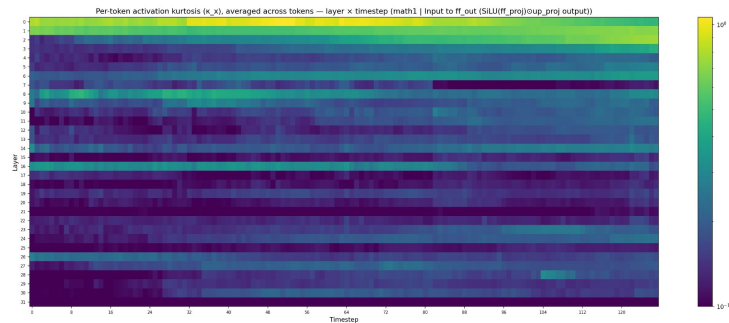
Unrotated Activations



Walsh-Hadamard Rotated (QuaRot)

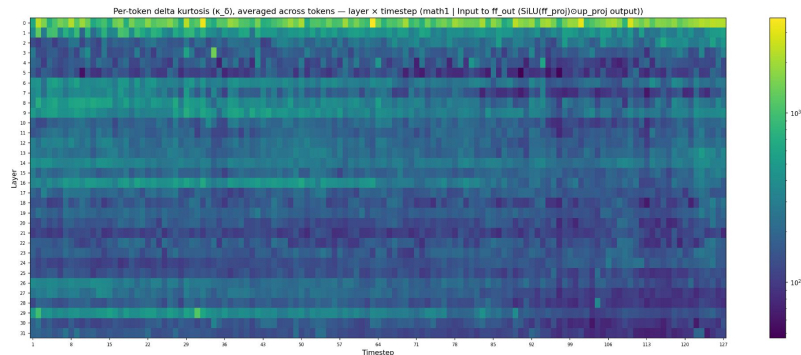


Block-Diagonal Rotated + Permuted (DuQuant)

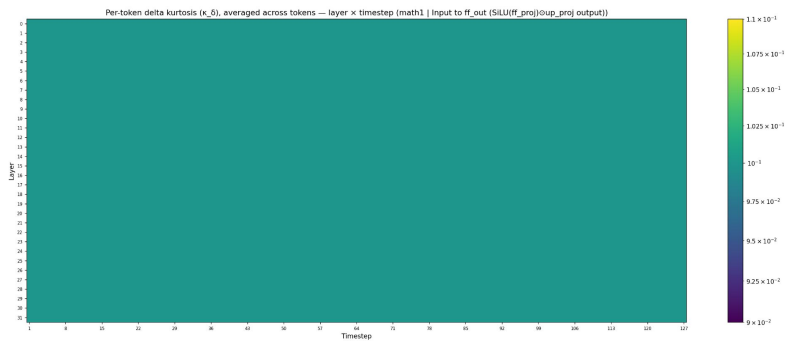


Delta Kurtosis : ff_out_in activations

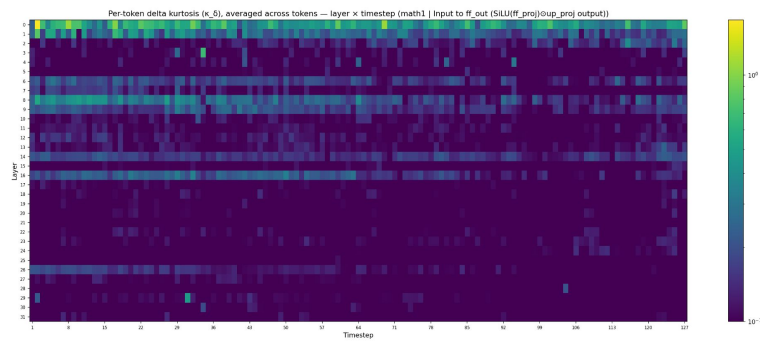
Unrotated Activations



Walsh-Hadamard Rotated (QuaRot)

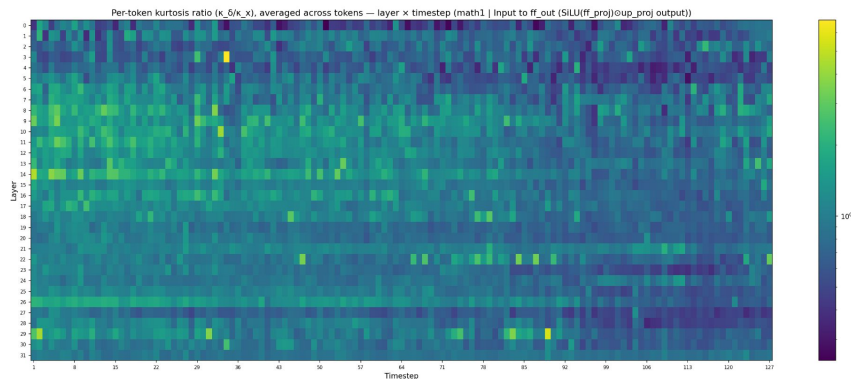


Block-Diagonal Rotated + Permuted (DuQuant)

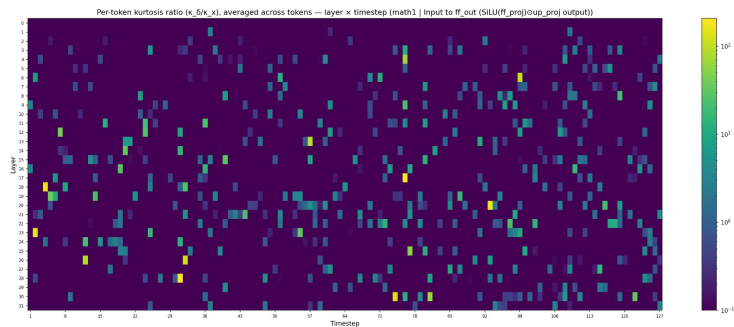


Kurtosis Ratio: ff_out_in activations

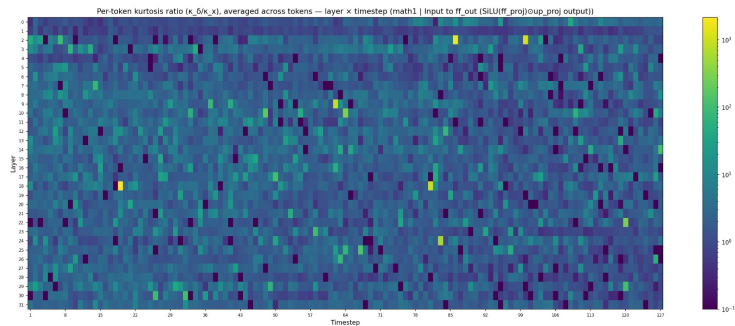
Unrotated Activations



Walsh-Hadamard Rotated (QuaRot)

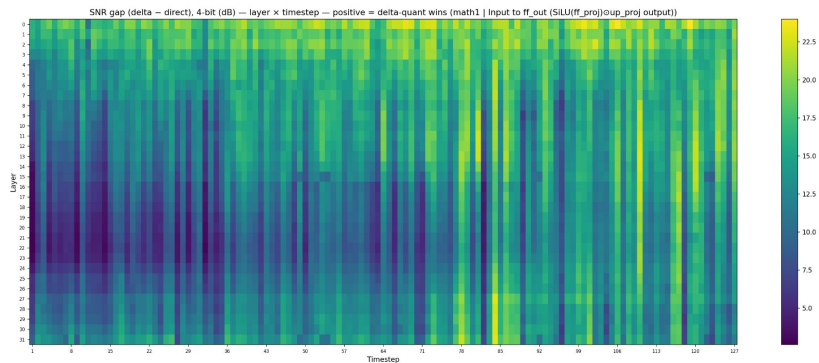


Block-Diagonal Rotated + Permuted (DuQuant)

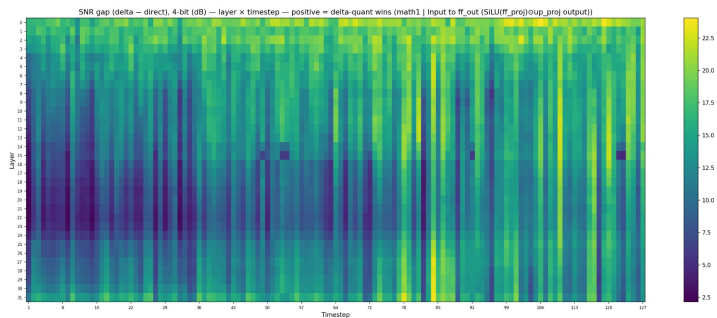


SNR Gap: ff_out_in activations

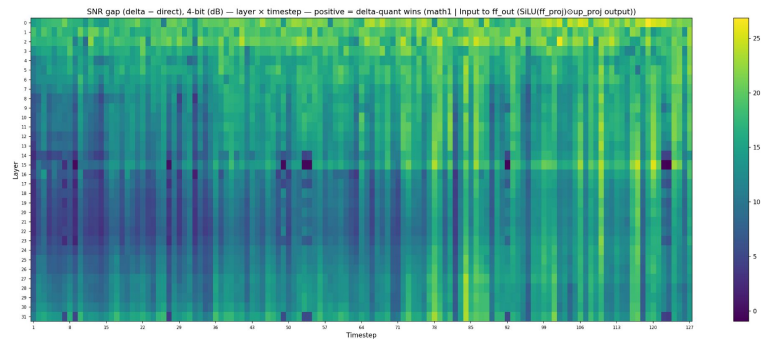
Unrotated Activations



Walsh-Hadamard Rotated (QuaRot)

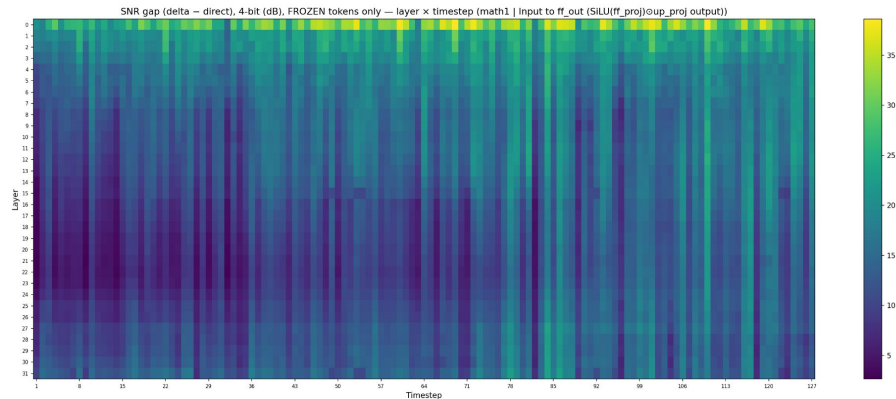


Block-Diagonal Rotated + Permuted (DuQuant)

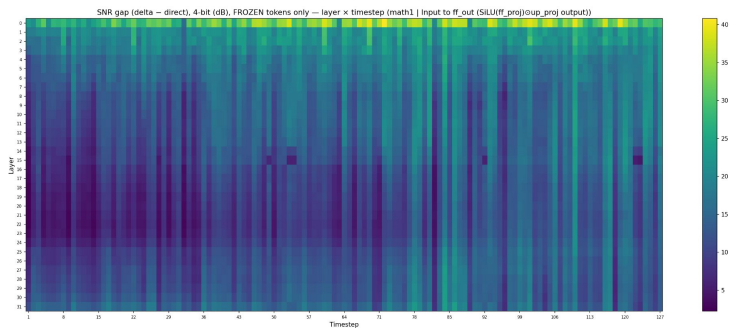


SNR Frozen Tokens Gap: ff_out_in activations

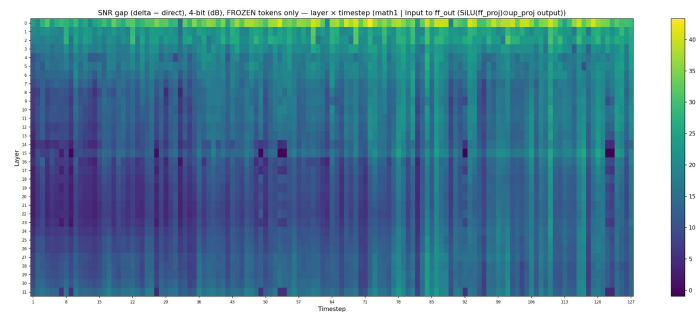
Unrotated Activations



Walsh-Hadamard Rotated (QuaRot)

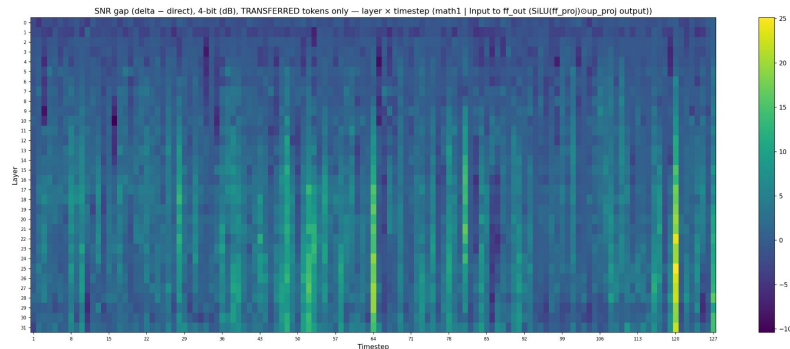


Block-Diagonal Rotated + Permuted (DuQuant)

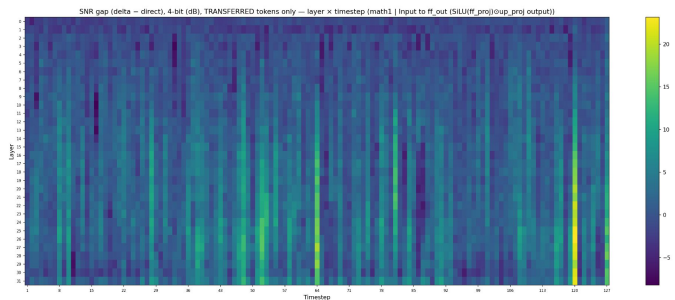


SNR Transferred Tokens Gap: ff_out_in activations

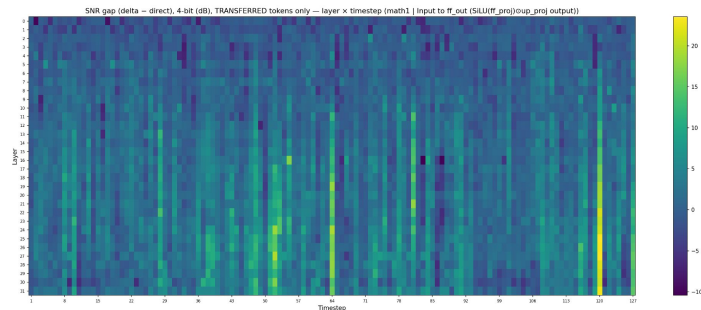
Unrotated Activations



Walsh-Hadamard Rotated (QuaRot)



Block-Diagonal Rotated + Permuted (DuQuant)



QuaRot Evaluation with 1 anchor

LLaDA-8B-Base accuracy under QuaRot rotation quantization setting. $\Delta\%$ is the relative change of the group average w.r.t. the FP16 base.

Model	Setting	Method	Math				Code			
			GSM8K	MATH	Avg	$\Delta\%$	HumanEval	MBPP	Avg	$\Delta\%$
LLaDA-8B-Base	FP16	Base	70.36	20.43	45.40	–	32.32	40.40	36.36	–
	W4A8	QuaRot	52.77	12.43	32.60	↓28.19%	22.56	32.60	27.58	↓24.1%
		DeltaQuaRot	51.93	11.14	31.54	↓30.53%	21.95	31.40	26.68	↓26.60%
	W4A4	QuaRot	37.68	9.57	23.63	↓47.95%	17.68	26.6	22.14	↓39.10%
		DeltaQuaRot	33.43	5.29	19.36	↓57.40%	14.63	22.60	18.62	↓48.79%

DuQuant Evaluation with 1 anchor

LLaDA-8B-Base accuracy under DuQuant rotation quantization setting. $\Delta\%$ is the relative change of the group average w.r.t. the FP16 base.

Model	Setting	Method	Math				Code			
			GSM8K	MATH	Avg	$\Delta\%$	HumanEval	MBPP	Avg	$\Delta\%$
LLaDA-8B-Base	FP16	Base	70.36	20.43	45.40	–	32.32	40.40	36.36	–
	W4A8	DuQuant	64.82	15.29	40.06	$\downarrow 11.76\%$	29.88	35.20	32.54	$\downarrow 10.50\%$
		DeltaDuQuant	64.75	17.14	40.95	$\downarrow 9.80\%$	31.71	35.20	33.46	$\downarrow 7.98\%$
	W4A4	DuQuant	59.67	16.43	38.05	$\downarrow 16.19\%$	24.39	32.00	28.20	$\downarrow 22.44\%$
		DeltaDuQuant	22.82	4.71	13.77	$\downarrow 69.67\%$	17.68	27.40	22.54	$\downarrow 38.01\%$

Delta Calibrated DuQuant

Original DuQuant is calibrated on raw activations from single forward pass which does not look at inter-timestep differences between activations

New Calibration Approach:

- Run the llada diffusion generation loop on wikitext2 prompts
- Compute difference between activations at consecutive timesteps
- Run the DuQuant rotation search method on the deltas

Delta Calibrated DuQuant

LLaDA-8B-Base accuracy under DuQuant rotation quantization settings. $\Delta\%$ is the relative change of the group average w.r.t. the FP16 base. [†]Rotation calibrated on inter-timestep deltas instead of raw activations.

Model	Setting	Method	Math				Code			
			GSM8K	MATH	Avg	$\Delta\%$	HumanEval	MBPP	Avg	$\Delta\%$
LLaDA-8B-Base	FP16	Base	70.36	20.43	45.40	–	32.32	40.40	36.36	–
	W4A8	DuQuant	64.82	15.29	40.06	↓ 11.76%	29.88	35.20	32.54	↓ 10.50%
		DeltaDuQuant	64.75	17.14	40.95	↓ 9.80%	31.71	35.20	33.46	↓ 7.98%
		DeltaDuQuant [†]	63.08	17.71	40.40	↓ 11.02%	31.71	36.20	33.96	↓ 6.61%
	W4A4	DuQuant	59.67	16.43	38.05	↓ 16.19%	24.39	32.00	28.20	↓ 22.44%
		DeltaDuQuant	22.82	4.71	13.77	↓ 69.67%	17.68	27.40	22.54	↓ 38.01%
		DeltaDuQuant [†]	28.43	6.57	17.50	↓ 61.45%	19.51	26.80	23.16	↓ 36.32%

Hyperparameters and Reanchoring Cost

Dataset	# fewshots	generation length	diffusion steps	block size
GSM8K	4	256	256	32
Math	0	256	256	64
HumanEval	0	512	512	32
MBPP	3	512	512	32

Table 37: Anchor interval (steps between FP16 cache resets, T/k) and FP16 fraction (k/T) for each dataset and anchor count k . GSM8K and Math share $T = 256$; HumanEval and MBPP share $T = 512$.

Anchors (k)	$T = 256$ (GSM8K, Math)		$T = 512$ (HumanEval, MBPP)	
	Interval (T/k)	FP16 %	Interval (T/k)	FP16 %
1	256	0.4	512	0.2
2	128	0.8	256	0.4
4	64	1.6	128	0.8
8	32	3.1	64	1.6
16	16	6.2	32	3.1
32	8	12.5	16	6.2

Uniform Re-anchoring Results

LLaDA-8B-Base accuracy under delta-quant on top of DuQuant/QuaRot rotations, swept over the number of full-precision anchors spread uniformly across the generation trajectory.

Model	Setting	Method	Anchors	Math				Code			
				GSM8K	MATH	Avg	$\Delta\%$	HumanEval	MBPP	Avg	$\Delta\%$
LLaDA-8B-Base	FP16	Base	–	70.36	20.43	45.40	–	32.32	40.40	36.36	–
			2	64.59	17.86	41.23	↓ 9.20%	31.71	35.20	33.46	↓ 7.99%
			4	64.44	17.71	41.08	↓ 9.53%	31.71	35.20	33.46	↓ 7.99%
			8	64.44	16.00	40.22	↓ 11.40%	32.32	35.00	33.66	↓ 7.41%
			16	65.81	16.71	41.26	↓ 9.07%	31.71	35.60	33.66	↓ 7.41%
	W4A8	DeltaDuQuant	32	65.96	16.14	41.05	↓ 9.58%	31.10	34.80	32.95	↓ 9.38%
			2	52.39	12.00	32.20	↓ 29.08%	21.95	31.40	26.68	↓ 26.62%
			4	52.54	11.71	32.13	↓ 29.23%	21.95	31.20	26.58	↓ 26.90%
			8	52.39	11.71	32.05	↓ 29.40%	21.95	31.20	26.58	↓ 26.90%
			16	52.99	12.43	32.71	↓ 27.95%	20.73	32.00	26.37	↓ 27.48%
		DeltaQuaRot	32	52.24	12.14	32.19	↓ 29.09%	21.34	31.20	26.27	↓ 27.75%
			2	24.26	5.57	14.92	↓ 67.14%	17.68	27.40	22.54	↓ 38.01%
			4	43.14	7.71	25.43	↓ 43.99%	17.68	27.40	22.54	↓ 38.01%
			8	53.53	13.14	33.34	↓ 26.57%	18.29	27.80	23.05	↓ 36.62%
			16	58.76	17.00	37.88	↓ 16.56%	22.56	30.20	26.38	↓ 27.45%
	W4A4	DeltaDuQuant	32	60.65	17.14	38.90	↓ 14.33%	24.39	31.80	28.10	↓ 22.73%
			2	33.06	5.29	19.18	↓ 57.76%	14.63	22.60	18.62	↓ 48.81%
			4	39.73	6.43	23.08	↓ 49.16%	15.24	22.80	19.02	↓ 47.69%
			8	45.49	9.14	27.32	↓ 39.83%	15.24	23.00	19.12	↓ 47.41%
			16	49.51	10.57	30.04	↓ 33.83%	15.24	24.40	19.82	↓ 45.49%
		DeltaQuaRot	32	51.18	11.29	31.24	↓ 31.20%	18.90	29.00	23.95	↓ 34.13%

Uniform Reanchoring Results

LLaDA-8B-Base accuracy under QuaRot rotation quantization setting. $\Delta\%$ is the relative change of the group average w.r.t. the FP16 base.

Model	Setting	Method	Math				Code			
			GSM8K	MATH	Avg	$\Delta\%$	HumanEval	MBPP	Avg	$\Delta\%$
LLaDA-8B-Base	FP16	Base	70.36	20.43	45.40	–	32.32	40.40	36.36	–
	W4A8	QuaRot	52.77	12.43	32.60	$\downarrow 28.19\%$	22.56	32.60	27.58	$\downarrow 24.10\%$
		DeltaQuaRot	51.93	11.14	31.54	$\downarrow 30.53\%$	21.95	31.40	26.68	$\downarrow 26.60\%$
		DeltaQuaRot (16 anch.)	52.99	12.43	32.71	$\downarrow 27.95\%$	20.73	32.00	26.37	$\downarrow 27.48\%$
	W4A4	QuaRot	37.68	9.57	23.63	$\downarrow 47.95\%$	17.68	26.60	22.14	$\downarrow 39.10\%$
		DeltaQuaRot	33.43	5.29	19.36	$\downarrow 57.40\%$	14.63	22.60	18.62	$\downarrow 48.79\%$
		DeltaQuaRot (32 anch.)	51.18	11.29	31.24	$\downarrow 31.20\%$	18.90	29.00	23.95	$\downarrow 34.13\%$

LLaDA-8B-Base accuracy under DuQuant rotation quantization setting. $\Delta\%$ is the relative change of the group average w.r.t. the FP16 base.

Model	Setting	Method	Math				Code			
			GSM8K	MATH	Avg	$\Delta\%$	HumanEval	MBPP	Avg	$\Delta\%$
LLaDA-8B-Base	FP16	Base	70.36	20.43	45.40	–	32.32	40.40	36.36	–
	W4A8	DuQuant	64.82	15.29	40.06	$\downarrow 11.76\%$	29.88	35.20	32.54	$\downarrow 10.50\%$
		DeltaDuQuant	64.75	17.14	40.95	$\downarrow 9.80\%$	31.71	35.20	33.46	$\downarrow 7.98\%$
		DeltaDuQuant (16 anch.)	65.81	16.71	41.26	$\downarrow 9.07\%$	31.71	35.60	33.66	$\downarrow 7.41\%$
	W4A4	DuQuant	59.67	16.43	38.05	$\downarrow 16.19\%$	24.39	32.00	28.20	$\downarrow 22.44\%$
		DeltaDuQuant	22.82	4.71	13.77	$\downarrow 69.67\%$	17.68	27.40	22.54	$\downarrow 38.01\%$
		DeltaDuQuant (32 anch.)	60.65	17.14	38.90	$\downarrow 14.33\%$	24.39	31.80	28.10	$\downarrow 22.73\%$

Block Re-anchoring Results

LLaDA-8B-Base accuracy: uniform reanchoring vs. block reanchoring under delta-quant on top of DuQuant/QuaRot rotations. $\Delta\%$ is the relative change of the group average w.r.t. the FP16 base. For block reanchoring the anchor count varies by dataset: GSM8K uses 8, MATH uses 4, HumanEval and MBPP use 16. The uniform rows use the matching per-dataset anchor count (GSM8K column: 8; MATH column: 4; code columns: 16).

Model	Setting	Method	Reanchor	Math				Code			
				GSM8K	MATH	Avg	$\Delta\%$	HumanEval	MBPP	Avg	$\Delta\%$
LLaDA-8B-Base	FP16	Base	–	70.36	20.43	45.40	–	32.32	40.40	36.36	–
	W4A8	DeltaDuQuant	uniform	64.44	17.71	41.08	↓9.52%	31.71	35.60	33.66	↓7.41%
				64.67	17.71	41.19	↓9.27%	31.71	35.60	33.66	↓7.43%
		DeltaQuaRot	uniform	52.39	11.71	32.05	↓29.40%	20.73	32.00	26.37	↓27.48%
				52.46	11.71	32.09	↓29.32%	20.73	32.00	26.37	↓27.48%
	W4A4	DeltaDuQuant	uniform	53.53	7.71	30.62	↓32.55%	22.56	30.20	26.38	↓27.45%
				54.74	7.71	31.23	↓31.21%	22.56	30.20	26.38	↓27.45%
		DeltaQuaRot	uniform	49.51	6.43	27.97	↓38.41%	15.24	24.40	19.82	↓45.49%
				45.49	6.43	25.96	↓42.82%	15.24	24.40	19.82	↓45.49%

Clipping Experiments (still in progress)

Table 36: Effect of clipping on LLaDA-8B-Base accuracy results

Model	Setting	Method	Clip	Math				Code			
				GSM8K	MATH	Avg	$\Delta\%$	HumanEval	MBPP	Avg	$\Delta\%$
LLaDA-8B-Base	FP16	Base	–	70.36	20.43	45.40	–	32.32	40.40	36.36	–
	W4A8	DeltaDuQuant	8 anchors	64.44	16.00	40.22	$\downarrow 11.40\%$	32.32	35.00	33.66	$\downarrow 7.41\%$
			0.9999	–	14.71	–	–	31.10	–	–	–
			0.999	–	15.71	–	–	32.32	–	–	–
			0.99	–	18.29	–	–	30.49	–	–	–
		DeltaQuaRot	8 anchors	52.39	11.71	32.05	$\downarrow 29.40\%$	21.95	31.20	26.58	$\downarrow 26.90\%$
			0.9999	–	12.14	–	–	22.56	–	–	–
			0.999	–	13.14	–	–	22.56	–	–	–
			0.99	–	12.71	–	–	21.95	–	–	–
	W4A4	DeltaDuQuant	8 anchors	53.53	13.14	33.34	$\downarrow 26.57\%$	18.29	27.80	23.05	$\downarrow 36.62\%$
			0.9999	–	12.71	–	–	20.73	–	–	–
			0.999	–	16.00	–	–	20.73	–	–	–
			0.99	–	17.00	–	–	23.78	–	–	–
		DeltaQuaRot	8 anchors	45.49	9.14	27.32	$\downarrow 39.83\%$	15.24	23.00	19.12	$\downarrow 47.41\%$
			0.9999	–	8.86	–	–	15.24	–	–	–
			0.999	–	10.29	–	–	17.68	–	–	–
			0.99	–	10.71	–	–	18.90	–	–	–

Related Work

DLLMQuant: Quantizing Diffusion-based Large Language Models

Model	Method	Truth.	Arc.	Hella.	Wino.	PIQA	MMLU	C-EVAL	Hum.	GSM8K	Avg.
LLADA	FP	47.49	44.03	54.06	74.9	74.65	65.85	69.54	32.92	67.48	59.87
	RTN	40.45	41.83	45.40	64.72	67.95	49.26	57.95	14.02	16.56	44.23
	AWQ	40.87	42.92	46.14	66.88	69.43	51.22	58.43	20.10	36.88	48.09
	textDLLMQuant ⁺	41.53	43.44	46.51	67.87	70.12	51.72	59.38	22.13	40.66	49.26
	QuaRot	42.53	44.20	49.76	69.85	70.75	55.96	56.32	25.33	44.57	51.03
	textDLLMQuant ⁺⁺	43.53	44.18	51.00	71.85	73.94	57.77	61.22	28.92	56.25	54.29
LLADA-1.5	FP	47.2	88.5	74.7	74.8	74.86	66.0	70.05	49.4	83.3	69.86
	RTN	39.51	81.77	56.82	65.27	66.91	48.88	58.96	23.44	36.56	53.12
	AWQ	40.96	82.22	66.84	67.93	69.32	51.12	60.03	30.07	57.95	58.49
	textDLLMQuant ⁺	42.14	83.38	70.09	68.69	70.22	51.63	61.28	32.44	59.55	59.94
	QuaRot	43.21	84.23	65.34	69.55	70.17	56.23	57.66	37.33	65.86	61.06
	DLLMQuant ⁺⁺	43.87	84.18	69.20	71.74	73.58	57.27	60.04	44.58	74.33	64.31
DREAM	FP	49.76	59.80	73.30	74.50	75.66	69.5	64.89	57.9	77.2	66.94
	RTN	41.25	54.34	58.32	64.52	66.26	51.51	49.89	28.90	30.82	49.53
	AWQ	43.66	57.82	65.57	67.13	68.96	55.5	53.27	33.14	48.98	54.89
	textDLLMQuant ⁺	44.14	58.38	66.93	68.66	69.53	57.63	54.21	35.12	51.14	56.19
	QuaRot	47.58	58.18	67.13	70.05	70.36	69.5	53.19	34.48	59.20	58.85
	textDLLMQuant ⁺⁺	47.86	59.43	70.14	71.54	72.09	69.5	55.89	44.50	66.17	61.90

Table 1: Results of RTN, AWQ, QuaRot, and ours DLLMQuant with 4-bit weight and activation quantization among 9 tasks on LLADA-8B, LLADA-1.5-8B, DREAM-7B) . DLLMQuant⁺ denotes DLLMQuant based on AWQ, and DLLMQuant⁺⁺ denotes DLLMQuant based on QuaRot.

FreeAct: Freeing Activations for LLM Quantization

Manyi Zhang, Ji-Fu Li, Xianzhi Yu

	dLLMs			
	HumanEval	GSM8K	Math500	Avg.
	<i>LLaDA</i>			
16-bit Baseline	43.90	76.80	38.40	53.03
RTN (W8A8)	39.02	77.56	37.00	51.19
RTN (W4A4)	00.00	00.00	00.14	00.05
SmoothQuant	00.00	00.00	00.00	00.00
QuaRot	28.66	65.81	24.20	39.56
FlatQuant	39.63	74.37	33.60	49.20
FreeAct (ours)	42.68	75.74	34.60	51.01
	<i>Dream</i>			
16-bit Baseline	56.10	68.16	42.20	55.49
RTN (W8A8)	48.78	68.76	40.80	52.78
RTN (W4A4)	00.00	00.53	0.018	00.18
SmoothQuant	00.00	00.00	0.014	00.05
QuaRot	39.02	56.56	25.80	40.46
FlatQuant	46.34	63.53	34.60	48.16
FreeAct (ours)	50.00	68.84	32.60	50.48

Plan for next week

- Finish the clipping experiment and test how clipping helps at other reanchoring levels
- Compare the results with other related work
- Collect results for Math500, because it would have less variance than Math100